

GABRIEL PEREZ

Materials Engineer | Data Scientist

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ABOUT ME

Materials Engineer transitioning into Data, with experience in Computational Simulation in the industrial sector, using computational modeling and field data to analyze structural integrity problems and support maintenance decisions. Seeking my first professional opportunity in Data Science as a Data Scientist, applying programming, statistics, and Machine Learning to solve problems and support data-driven decision-making.

STACK

- Programming languages: Python, VBA and SQL.
- Libraries: Pandas, Numpy and Scikit-Learn
- Tools: Excel, Git and SQLite3

PROFESSIONAL EXPERIENCE

MAR. 2025 - PRESENT | PRINER GROUP

Computational Simulation Engineer

Responsible for conducting structural analyses to support decision-making in maintenance and inspection activities.

- Performed numerical analyses using the Finite Element Method to assess equipment integrity, integrating design information and historical maintenance data retrieved from CMMS systems to support inspection and maintenance decisions.
- Conducted fitness-for-service assessments of equipment affected by corrosion-induced thickness loss and crack-like defects, contributing to the extension of equipment service life.
- Developed Risk-Based Inspection (RBI) plans for offshore platforms by analyzing and organizing design data, such as piping, material, and valve lists, supporting the implementation of proactive maintenance and inspection strategies.

DEZ. 2023 - DEZ. 2024 | THEMAG-IDOM-MAGNA CONSORTIUM

Piping Flexibility Designer

Supported piping modeling and structural analysis activities for the electromechanical design of the Angra 3 Nuclear Power Plant.

- Developed a Python automation script using Selenium to automate the collection and organization of files made available by the client through SharePoint, eliminating manual information-gathering tasks.
- Developed a VBA tool for post-processing numerical analysis results of hyperstatic supports, automating the verification of specific criteria and streamlining results evaluation.

JUL. 2023 - DEZ. 2023 | TRESKA ENGENHARIA

Computational Simulation Intern

Prepared computational models for compliance with engineering standards, supporting inspection and maintenance activities.

FEV. 2022 - JUL. 2023 | BRASKEM Q3

Static Equipment Inspection Intern

I worked in equipment inspection at a petrochemical plant, structuring data for audits and maintenance analyses. I automated data integration into IBM Maximo using Python and Selenium, reducing the time required to make information available for brittle fracture analyses from months to just a few days. I also developed a Python application to automate temperature analyses and report generation.

EDUCATION

2025 - 2026 | FRONS EDUCAÇÃO

Postgraduate Certificate in Structural Analysis and Finite Element Analysis

2018 - 2025 | UNIVERSIDADE FEDERAL DO ABC (UFABC)

Bachelor of Science and Technology

Bachelor of Materials Engineering

LANGUAGES

- English (B2 certified)
- Portuguese (native)